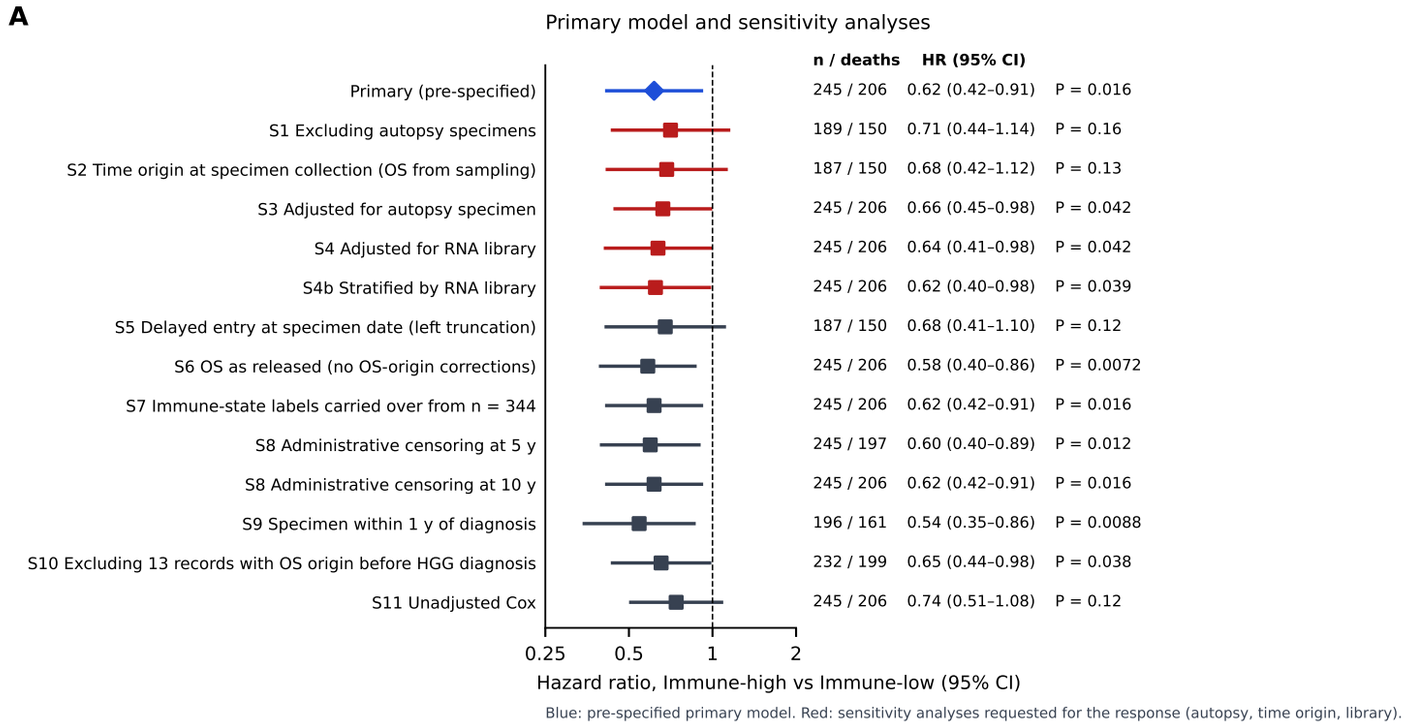
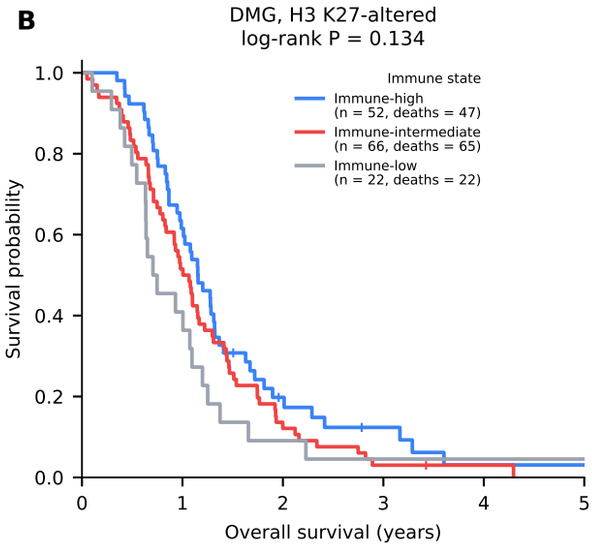
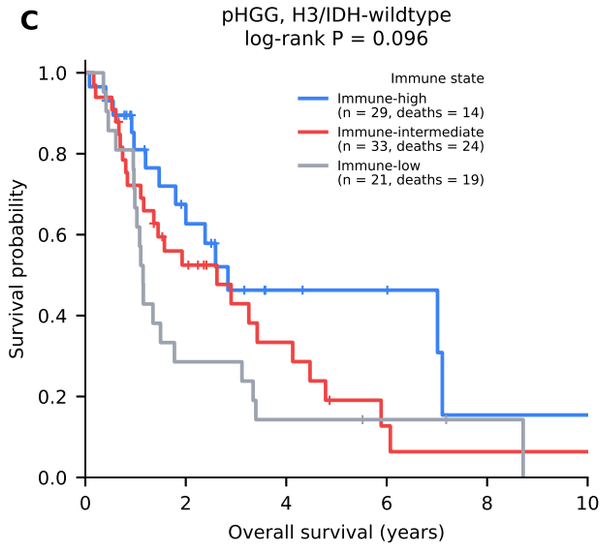
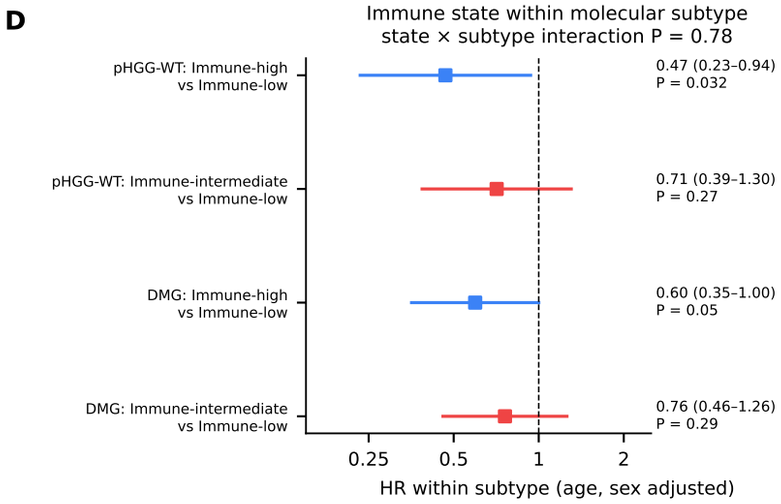
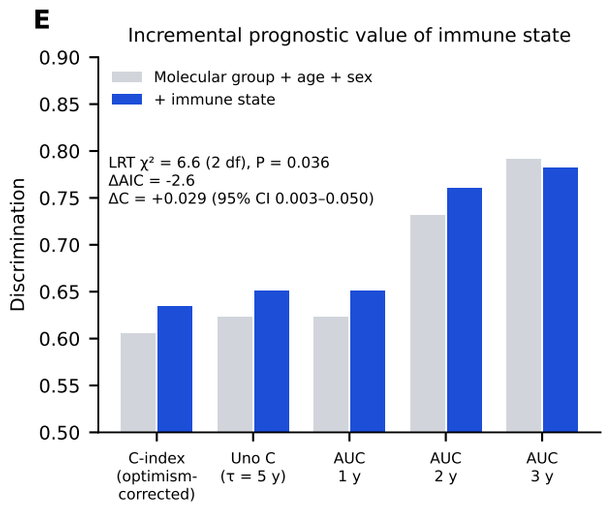

A
Primary model and sensitivity analyses
n / deaths
HR (95% CI)
Primary (pre-specified) 245 / 206 0.62 (0.42–0.91) P = 0.016
S1 Excluding autopsy specimens 189 / 150 0.71 (0.44–1.14) P = 0.16
S2 Time origin at specimen collection (OS from sampling) 187 / 150 0.68 (0.42–1.12) P = 0.13
S3 Adjusted for autopsy specimen 245 / 206 0.66 (0.45–0.98) P = 0.042
S4 Adjusted for RNA library 245 / 206 0.64 (0.41–0.98) P = 0.042
S4b Stratified by RNA library 245 / 206 0.62 (0.40–0.98) P = 0.039
S5 Delayed entry at specimen date (left truncation) 187 / 150 0.68 (0.41–1.10) P = 0.12
S6 OS as released (no OS-origin corrections) 245 / 206 0.58 (0.40–0.86) P = 0.0072
S7 Immune-state labels carried over from n = 344 245 / 206 0.62 (0.42–0.91) P = 0.016
S8 Administrative censoring at 5 y 245 / 197 0.60 (0.40–0.89) P = 0.012
S8 Administrative censoring at 10 y 245 / 206 0.62 (0.42–0.91) P = 0.016
S9 Specimen within 1 y of diagnosis 196 / 161 0.54 (0.35–0.86) P = 0.0088
S10 Excluding 13 records with OS origin before HGG diagnosis 232 / 199 0.65 (0.44–0.98) P = 0.038
S11 Unadjusted Cox 245 / 206 0.74 (0.51–1.08) P = 0.12
0.25
0.5
1
2
Hazard ratio, Immune-high vs Immune-low (95% CI)
Blue: pre-specified primary model. Red: sensitivity analyses requested for the response (autopsy, time origin, library).


B
DMG, H3 K27-altered
log-rank P = 0.134
Immune state
Immune-high (n = 52, deaths = 47)
Immune-intermediate (n = 66, deaths = 65)
Immune-low (n = 22, deaths = 22)
Survival probability
1.0
0.8
0.6
0.4
0.2
0.0
0
1
2
3
4
5
Overall survival (years)
At risk
Immune-high 52 32 8 4 1 1
Immune-intermediate 66 34 8 2 1 0
Immune-low 22 9 2 1 1 1


C
pHGG, H3/IDH-wildtype
log-rank P = 0.096
Immune state
Immune-high (n = 29, deaths = 14)
Immune-intermediate (n = 33, deaths = 24)
Immune-low (n = 21, deaths = 19)
Survival probability
1.0
0.8
0.6
0.4
0.2
0.0
0
2
4
6
8
10
Overall survival (years)
At risk
Immune-high 29 14 5 4 1 1
Immune-intermediate 33 15 7 2 1 1
Immune-low 21 6 3 2 1 0


D
Immune state within molecular subtype
state × subtype interaction P = 0.78
pHGG-WT: Immune-high vs Immune-low
0.47 (0.23–0.94) P = 0.032
pHGG-WT: Immune-intermediate vs Immune-low
0.71 (0.39–1.30) P = 0.27
DMG: Immune-high vs Immune-low
0.60 (0.35–1.00) P = 0.05
DMG: Immune-intermediate vs Immune-low
0.76 (0.46–1.26) P = 0.29
0.25
0.5
1
2
HR within subtype (age, sex adjusted)


E
Incremental prognostic value of immune state
Molecular group + age + sex
+ immune state
LRT χ² = 6.6 (2 df), P = 0.036
ΔAIC = -2.6
ΔC = +0.029 (95% CI 0.003–0.050)
Discrimination
0.90
0.85
0.80
0.75
0.70
0.65
0.60
0.55
0.50
C-index (optimism-corrected)
Uno C (τ = 5 y)
AUC 1 y
AUC 2 y
AUC 3 y